

Aanchal Rathi

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Summary

AI/ML student with experience building LLM-powered applications, RAG pipelines, and scalable backend systems. Skilled in Python, FastAPI, Vector databases, cloud deployment, and machine learning with a strong focus on production-ready AI solutions.

Education

VIT Bhopal University — B.Tech, Computer Science (AI & ML Specialization) *Sep 2023 – Present*
CGPA: 8.98 / 10

DAV Public School — 12th Grade: 83.8% | 10th Grade: 94.1% *2019 – 2023*

Technical Skills

- **Languages:** Python, C++, Java, SQL
- **AI/ML:** RAG, LLM APIs (Groq, Llama 3.3), Cohere Embeddings, Prompt Engineering, Scikit-learn, TF-IDF, Cosine Similarity, Pandas, NumPy, OpenCV
- **Backend & Databases:** FastAPI, Flask, REST APIs, Async Background Tasks, LangChain, ChromaDB, MongoDB Atlas, PostgreSQL, MySQL, Firebase, SQLAlchemy
- **Cloud & Tools:** AWS, Microsoft Azure (Certified), Docker, Render, Git/GitHub
- **Core:** DSA, OOP, DBMS, Operating Systems, Machine Learning, System Design Fundamentals

Projects

- **SEO Blog Agent – AI-Powered RAG Content Generation Platform** *May 2026 – Jun 2026*
FastAPI, Streamlit, PostgreSQL, ChromaDB, Cohere Embeddings, Gemini 2.5 Flash, Docker
 - Built an end-to-end RAG pipeline that retrieves brand-specific context from company documents using ChromaDB and Cohere embeddings, grounding LLM-generated content in real, verifiable data.
 - Designed an async job-queue architecture with FastAPI BackgroundTasks, enabling 30–90 second AI pipeline runs without blocking requests or hitting platform timeout limits.
 - Engineered an intent-based keyword scoring system using live Google Autocomplete and Serper.dev SERP data, with rule-based content-format selection across 6 distinct blog structures.
 - Diagnosed and resolved production-grade issues including memory-constrained OOM crashes and database connection pooling failures, deploying a fully containerized two-service architecture on Render.
- **Youth News Hub – Personalized News Recommendation Platform** *Mar 2026 – May 2026*
Python, Flask, JavaScript, MongoDB Atlas, Firebase Authentication, TF-IDF
 - Built a full-stack personalized news platform integrating multiple news APIs with automatic fallback handling and a TF-IDF + cosine similarity recommendation engine that dynamically ranks 15+ articles based on user preferences.
 - Integrated 2 external news APIs with automatic fallback handling and backend caching to improve content reliability under API failures and rate limits.
 - Implemented Firebase Authentication for secure user sessions and MongoDB Atlas for scalable storage of user preferences and reading history.
- **Crop Recommendation System – Machine Learning** *Feb 2025 – Apr 2025*
Python, Pandas, NumPy, Scikit-learn
 - Trained and compared Decision Tree, KNN, Naive Bayes, and SVM classifiers on soil nutrient and climate data, achieving 95% accuracy with Decision Tree for crop recommendation.

Achievements & Certifications

- Solved 500+ DSA problems across LeetCode and GeeksforGeeks.
- Certifications: Microsoft Azure Data Fundamentals, IBM SkillsBuild – Getting Started with AI, Machine Learning with Python (Coursera).

Leadership & Extracurriculars

- Core team member, University Dance Club – organized and lead 9 events with a team of 15–25 members (2024–Present).
- Winner, Inter-College Women's Cricket Tournament (2024).